



Implementing Advance Windows Server on E-Agriculture

Group 2

Haikal raihan Hafidz

Mufid Fata Rifki

Reyandanu Fajri Daniswara

Faculty:

Ivan Firdaus, ST.

Class:

2CS2

**CEP CCIT FACULTY OF ENGINEERING
UNIVERSITY OF INDONESIA
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PROJECT INFORMATION

Project Title : Implementing Advance Windows Server on E- Agriculture
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Name of Faculty : Ivan Firdaus, ST.

Names of Administrator:

1. Haikal Raihan Hafidz
2. Mufid Fata Rifki
3. Reyandanu Fajri Daniswara

ACKNOWLEDGEMENT

Thank God Almighty for His help and blessings, which enabled us to complete this project successfully. We would also like to express our sincere gratitude to Mr. Ivan Firdaus, S.T. for his support, advice, and guidance throughout the course of this project. His insights made it easier for us to understand and complete the system we developed.

This paper presents how we configured and implemented several server components using the Windows Server platform, including DHCP, DNS, ADDS, GPO, NFS, Join Server, DHCP Failover, File Server Resource Manager (FSRM), and Distributed File System (DFS). We focused on understanding each feature and applying it to support the development of an E-Agriculture system.

We hope this project will be helpful and serve as a reference for others who are interested in learning more about network infrastructure and server configuration.

Depok, June 3rd 2025

Authors

SYSTEM ANALYSIS

This E-Agriculture project uses the Windows Server system to help build a better and more organized farming network. The system includes several important servers, such as DHCP, DNS, ADDS, GPO, NFS, Join Server, DHCP Failover, File Server Resource Manager (FSRM), and Distributed File System (DFS). These servers work together to make sure the network is stable, easy to manage, and always available when needed.

With this setup, smart farming tools like IoT devices can connect and work more smoothly. It also helps store and share data safely, manage users easily, and keep the system running even if one part has a problem. Overall, this project supports a more modern and efficient farming process by using technology to make work easier and more accurate.

Configuring DHCP Server

First, test the dhcp by looking if the client has been assigned an IP from the dhcp with the input `ipconfig` in terminal

```
Command Prompt
Microsoft Windows [Version 10.0.19045.5737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Reyandanu>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet1:

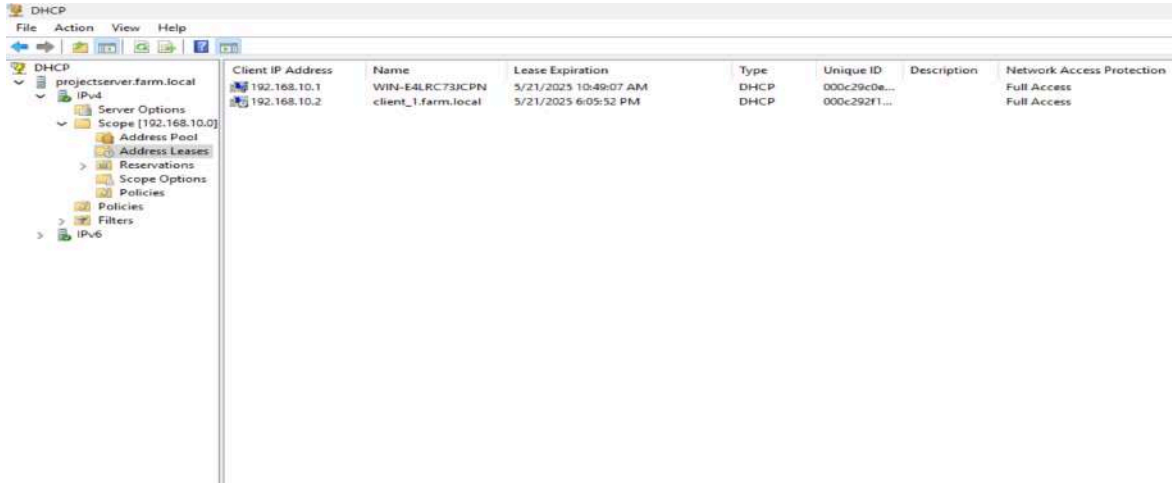
    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::7fcf:6707:25f3:f48f%12
    IPv4 Address. . . . . : 192.168.10.2
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.10.1

C:\Users\Reyandanu>
```

Next, look if the client shows up in the address pool option in DHCP Manager

Start IP Address	End IP Address	Description
192.168.10.1	192.168.10.200	Address range for distribution
192.168.10.10	192.168.10.20	IP Addresses excluded from distribution

To see the client's Ip leases, go to the Address Leases option in the DHCP Manager



Test the IP given by the DHCP by ping the server from the client PC

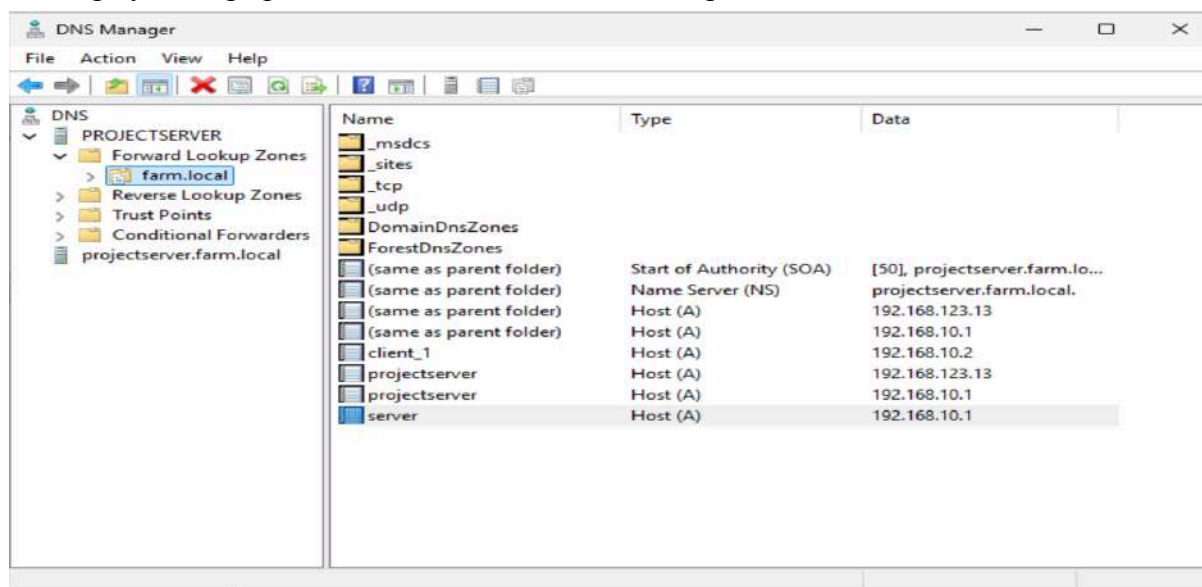
```
C:\Users\Reyandanu>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128

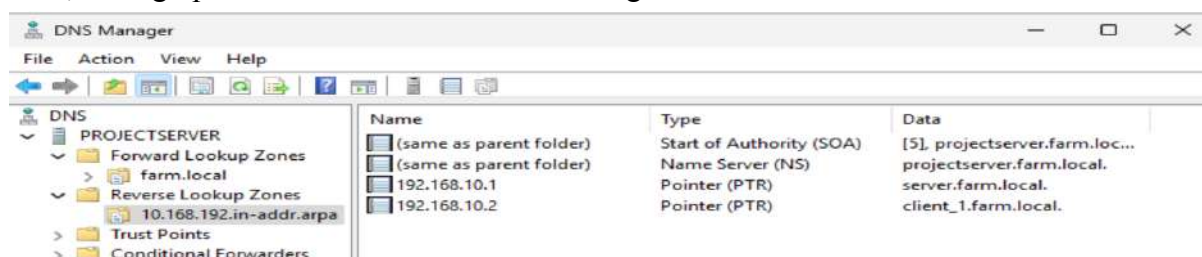
Ping statistics for 192.168.10.1:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Configuring DNS Server

Starting by setting up the Forward Zone in DNS Manager



Next, setting up the Reverse Zone in DNS Manager



Testing the DNS by pinging the DNS from the client PC

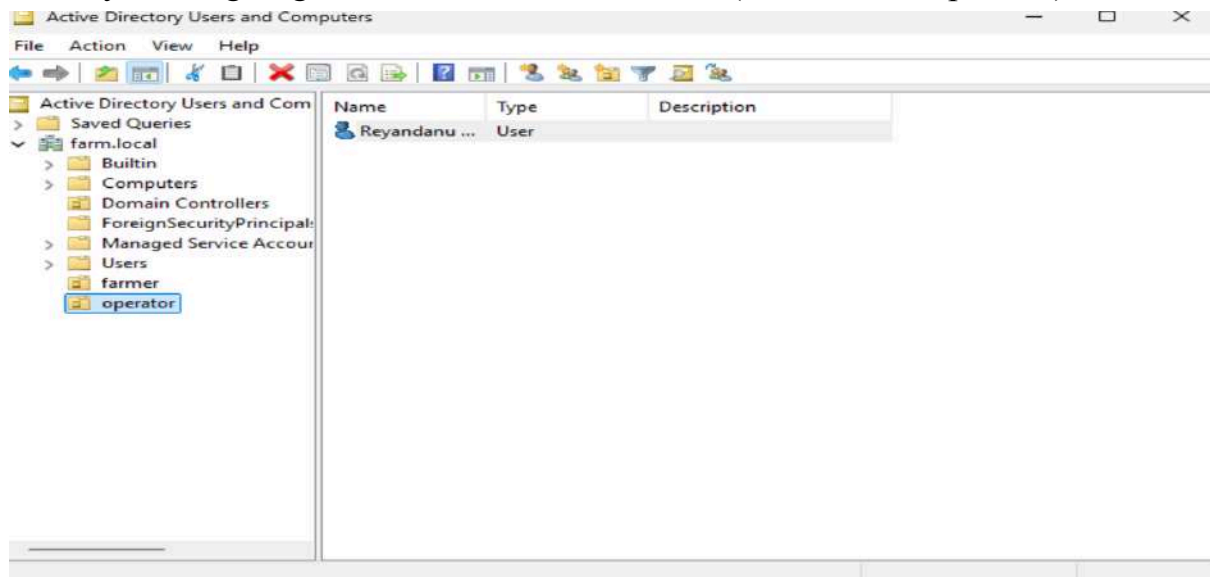
```
C:\Users\Reyandanu>ping farm.local

Pinging farm.local [192.168.10.1] with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128

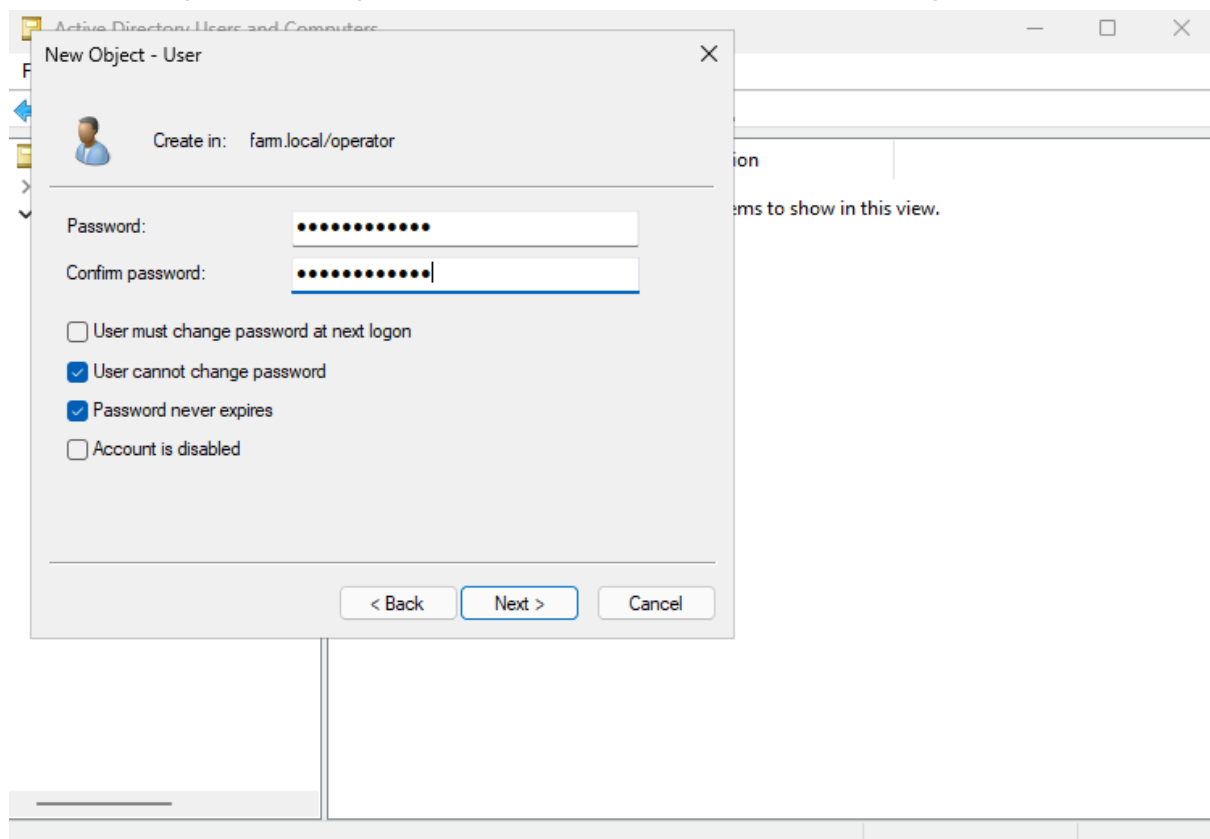
Ping statistics for 192.168.10.1:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Configuring ADDS Server

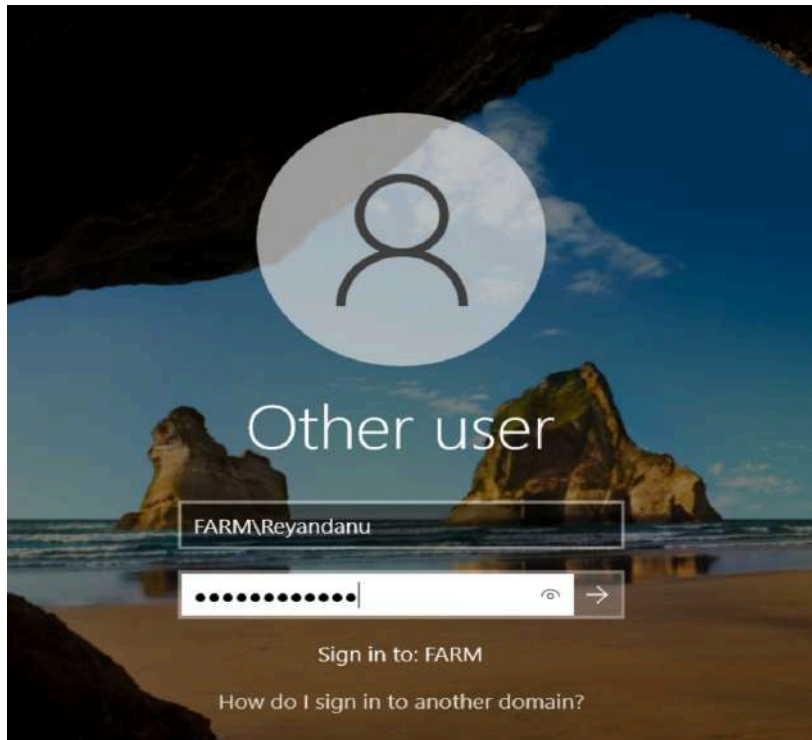
Start by making Organizational Units as needed (Farmer and Operator)



Next, adding and setting the user ADDS as needed on each organizational units



Testing the ADDS by signing in as a user in client PC



Configuring GPO

Setting up the policy as needed for the GPO, for example enabling the “Disable the Display Control Panel”, “Prohibit access to Control Panel and PC settings”, “Remove Run menu from the start”, and “Prevent access to the command prompt”.

Disable the Display Control Panel	Enabled
Prohibit access to Control Panel and PC settings	Enabled
Remove Run menu from Start Menu	Enabled
Prevent access to the command prompt	Enabled

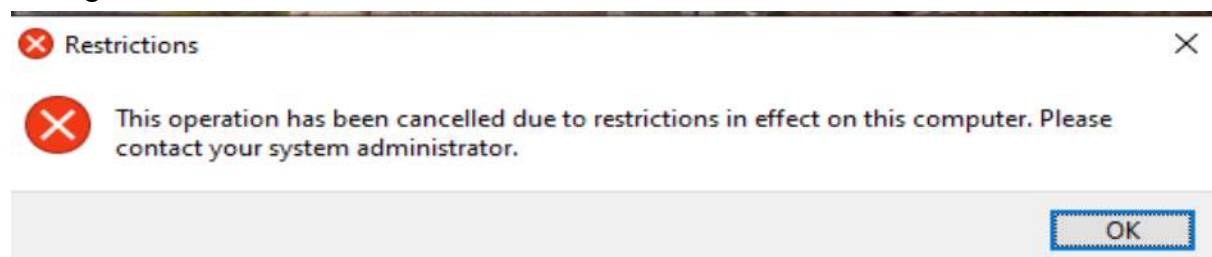
Next, testing if the GPO disabled cmd/terminl on client PC

```
CA Select Command Prompt
Microsoft Windows [Version 10.0.19045.5737]
(c) Microsoft Corporation. All rights reserved.

The command prompt has been disabled by your administrator.

Press any key to continue . . .
```

Testing if Control Panel is disabled on clients PC



Configuring NFS Server

Start with mounting the NFS file path on the server with the command `mount 192.168.10.1:/share i:\` and test if the client connected to the NFS

Command Prompt

```
Microsoft Windows [Version 10.0.19045.5737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Reyandanu>mount 192.168.10.1:/share i:\
i: is now successfully connected to 192.168.10.1:/share

The command completed successfully.

C:\Users\Reyandanu>
```

Check if the NFS is working on file manager such as below:



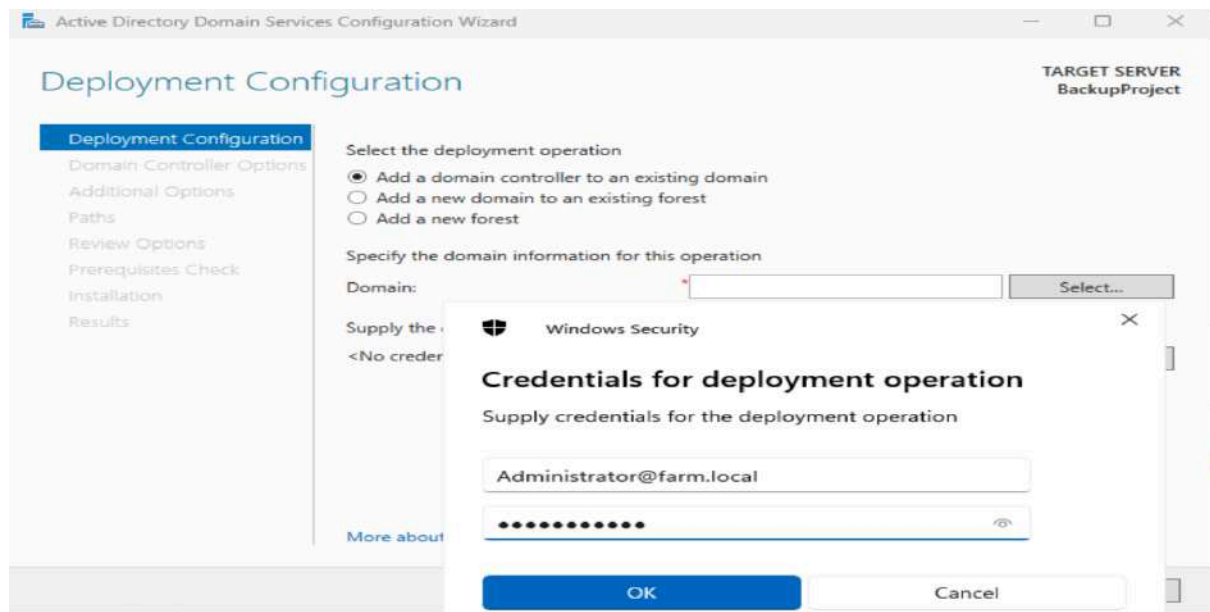
Configuring Join Forest

First, set the IP of the backup server to a static IP address but still on the same subnet.

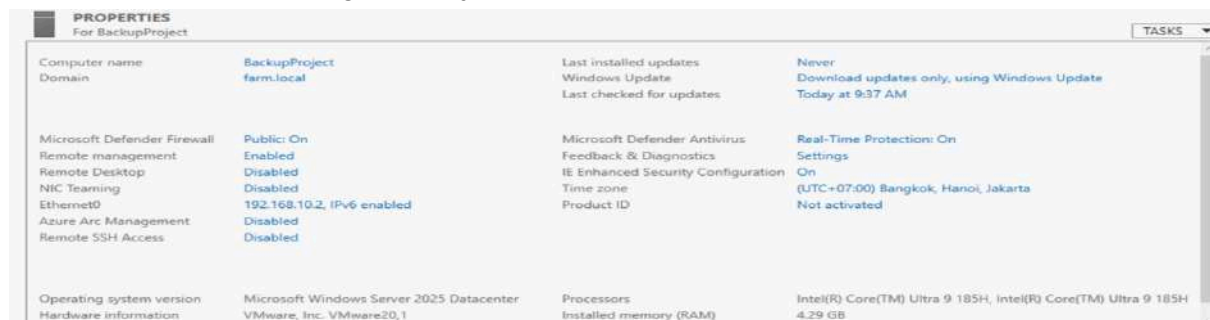
```
Ethernet adapter Ethernet0:

Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : fe80::fd9c:3ec3:3fd2:f614%6
IPv4 Address. . . . . : 192.168.10.2
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
```

Log in the backup server to the existing forest

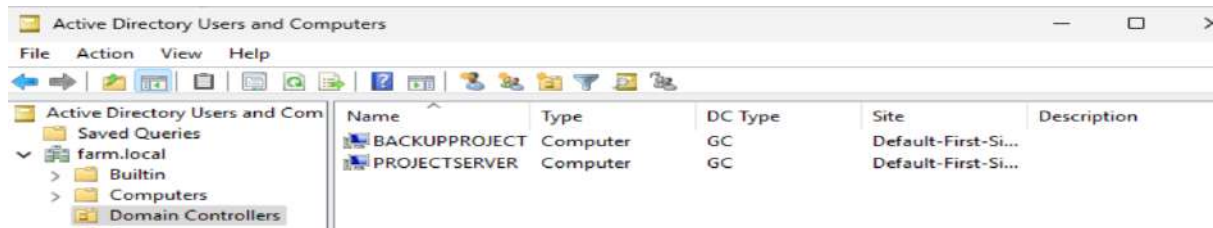


see the local server manager if the join forest success on the backup server

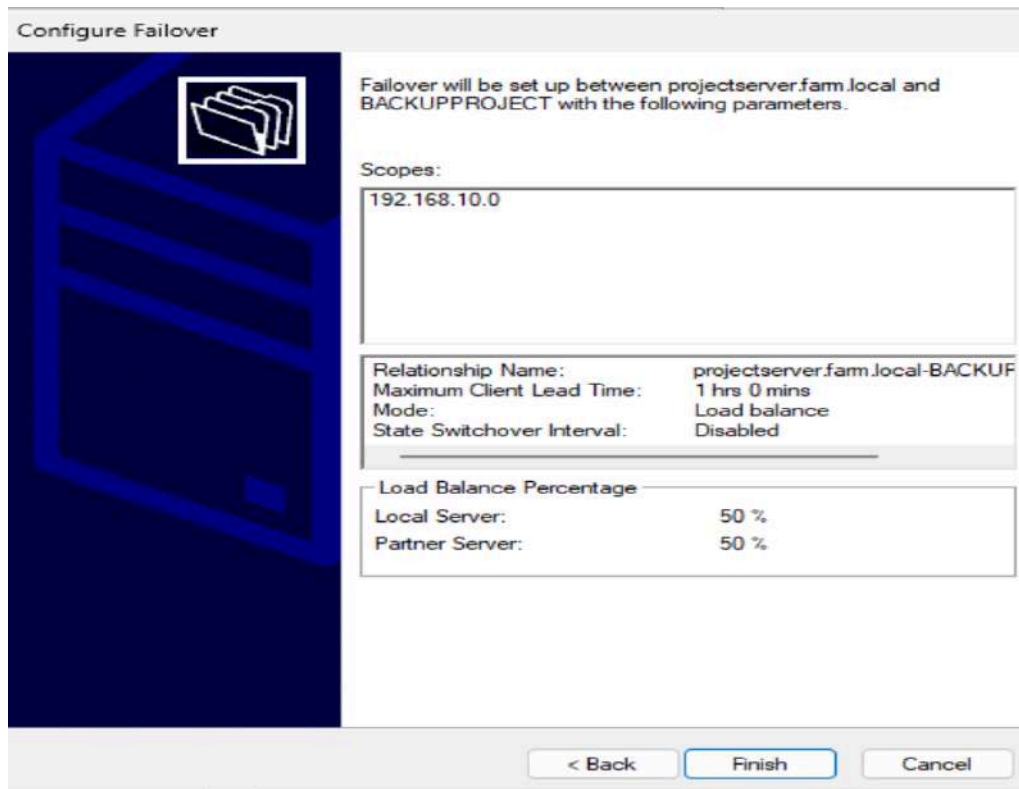


Configuring DHCP Failover

Check domain controllers menu in main server if the backup server showed up



Configure the DHCP Failover like below:



Test the DHCP Failover on client 192.168.10.1 (Main server turned off) and 192.168.10.2 (backup server)

```
PS C:\Users\Reyandanu> ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Control-C
PS C:\Users\Reyandanu> ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
Control-C
PS C:\Users\Reyandanu> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet1:

    Connection-specific DNS Suffix  . : farm.local
    Link-local IPv6 Address . . . . . : fe80::7fcf:6707:25f3:f48f%12
    IPv4 Address. . . . . : 192.168.10.5
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.10.2
PS C:\Users\Reyandanu>
```

Configuring FSRM

Make a new partition for FSRM on the main server

■ New Volume (E:)	Simple	Basic	NTFS	Healthy (Basic Data Partition)
■ Windows (C:)	Simple	Basic	NTFS	Healthy (Boot, Page File, Crash Dump, Basic Data Partition)

Create a File Screen Template for the farm

Create File Screen Template

Copy properties from template (optional):
Block Audio and Video Files

Settings | E-mail Message | Event Log | Command | Report

Template name:
Farm Screen

Screening type:
☒ Active screening: Do not allow users to save unauthorized files
☐ Passive screening: Allow users to save files (use for monitoring)

File groups
Select file groups to block:

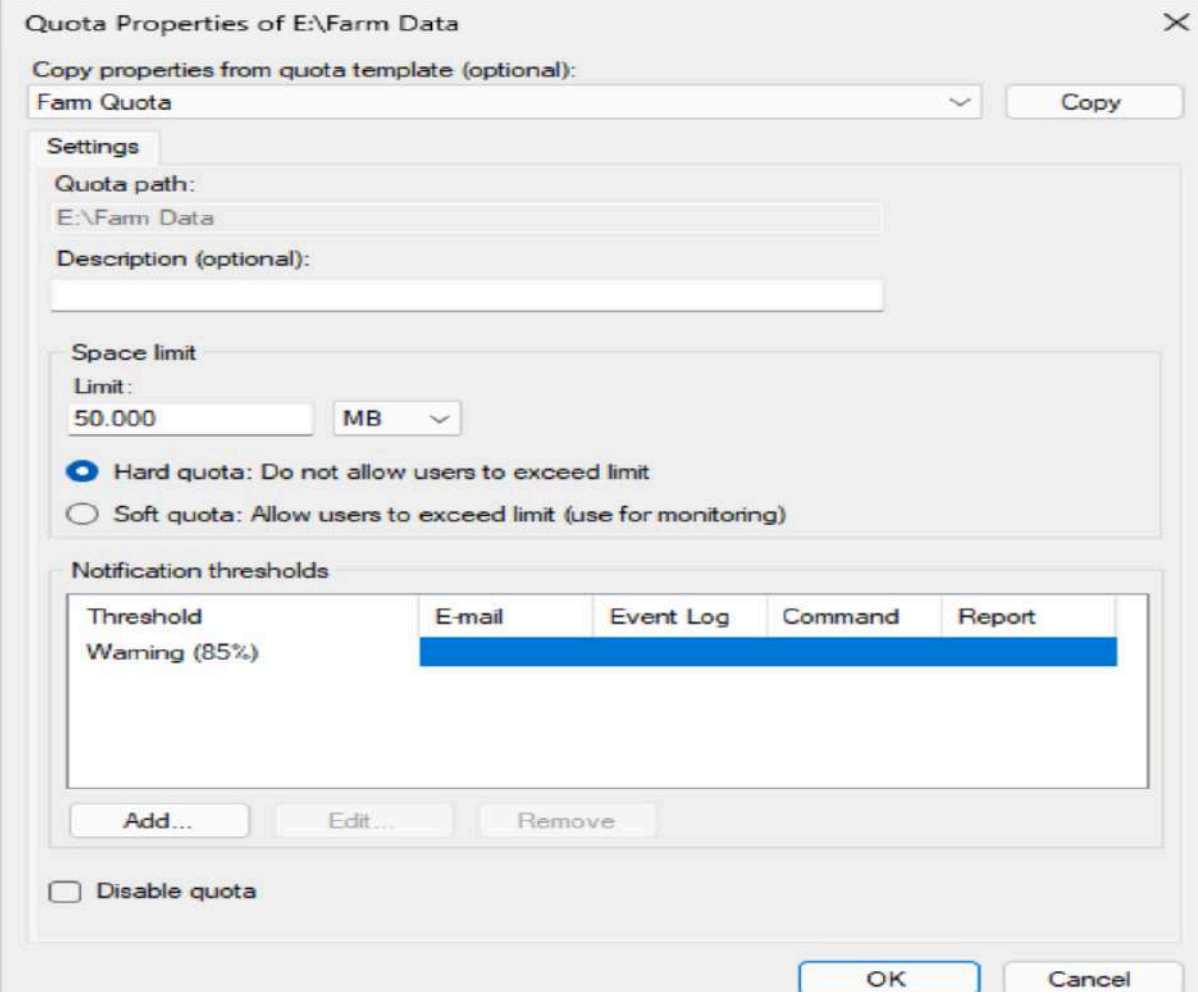
- ☐ E-mail Files
- ☐ Executable Files
- ☒ Image Files
- ☐ Office Files
- ☐ System Files
- ☐ Temporary Files
- ☒ Text Files
- ☐ Web Page Files

Maintain file groups:
Create...
Edit...

To select a file group to edit, highlight its label.

OK Cancel

Configuration of Quota properties for the FSRM



Quota Properties of E:\Farm Data

Copy properties from quota template (optional):
Farm Quota [v] [Copy]

Settings

Quota path:
E:\Farm Data

Description (optional):
[]

Space limit

Limit:
50.000 MB [v]

☒ Hard quota: Do not allow users to exceed limit
☐ Soft quota: Allow users to exceed limit (use for monitoring)

Notification thresholds

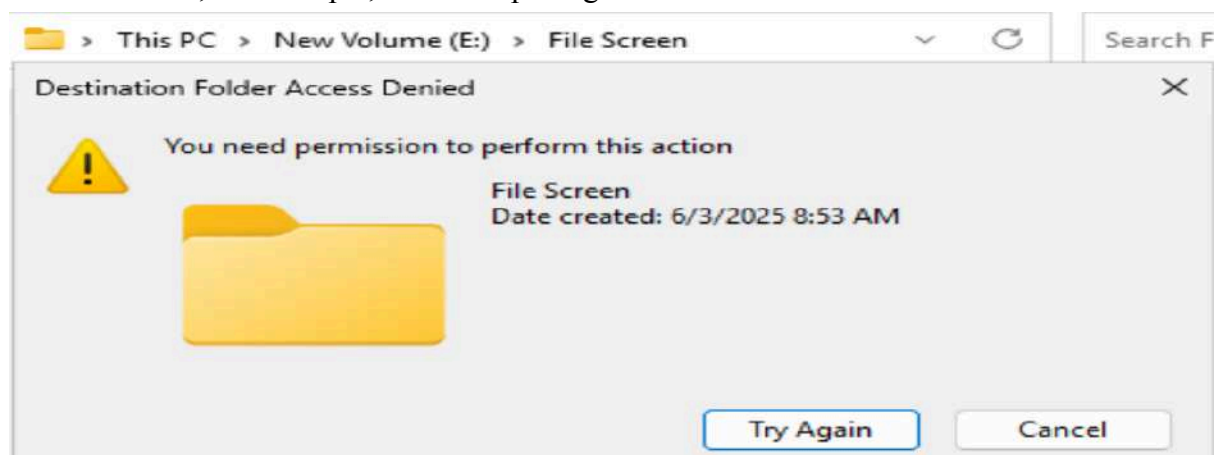
Threshold	E-mail	Event Log	Command	Report
Warning (85%)				

[Add...] [Edit...] [Remove]

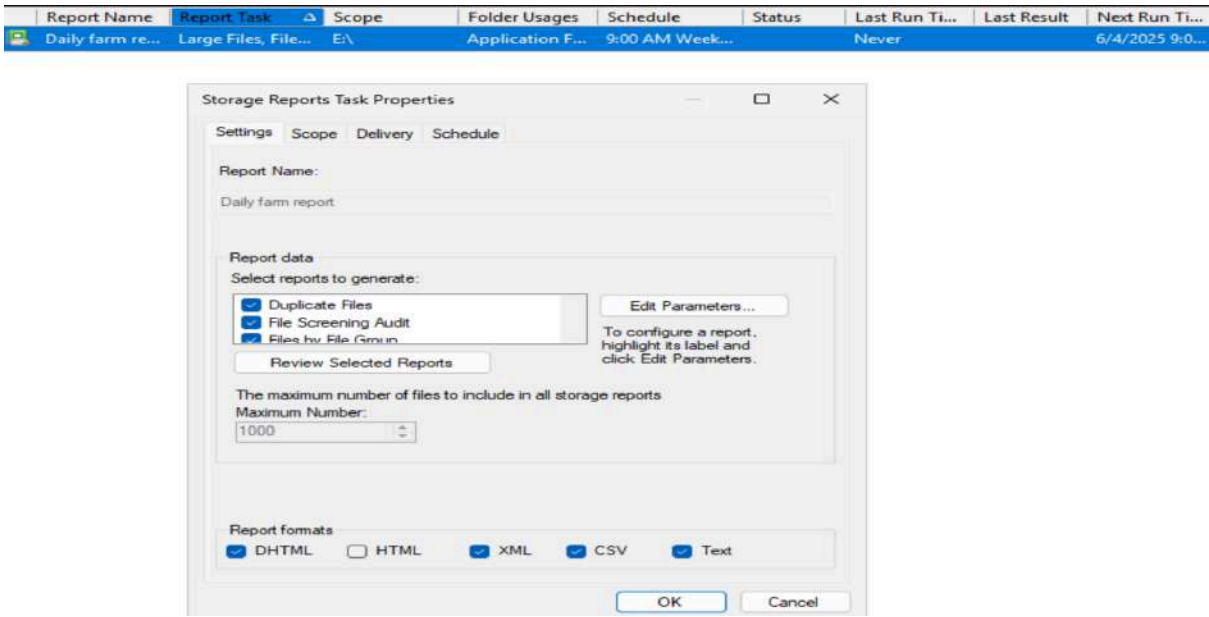
☐ Disable quota

[OK] [Cancel]

Test the FSRM, for example, this bitmap image on the file folder



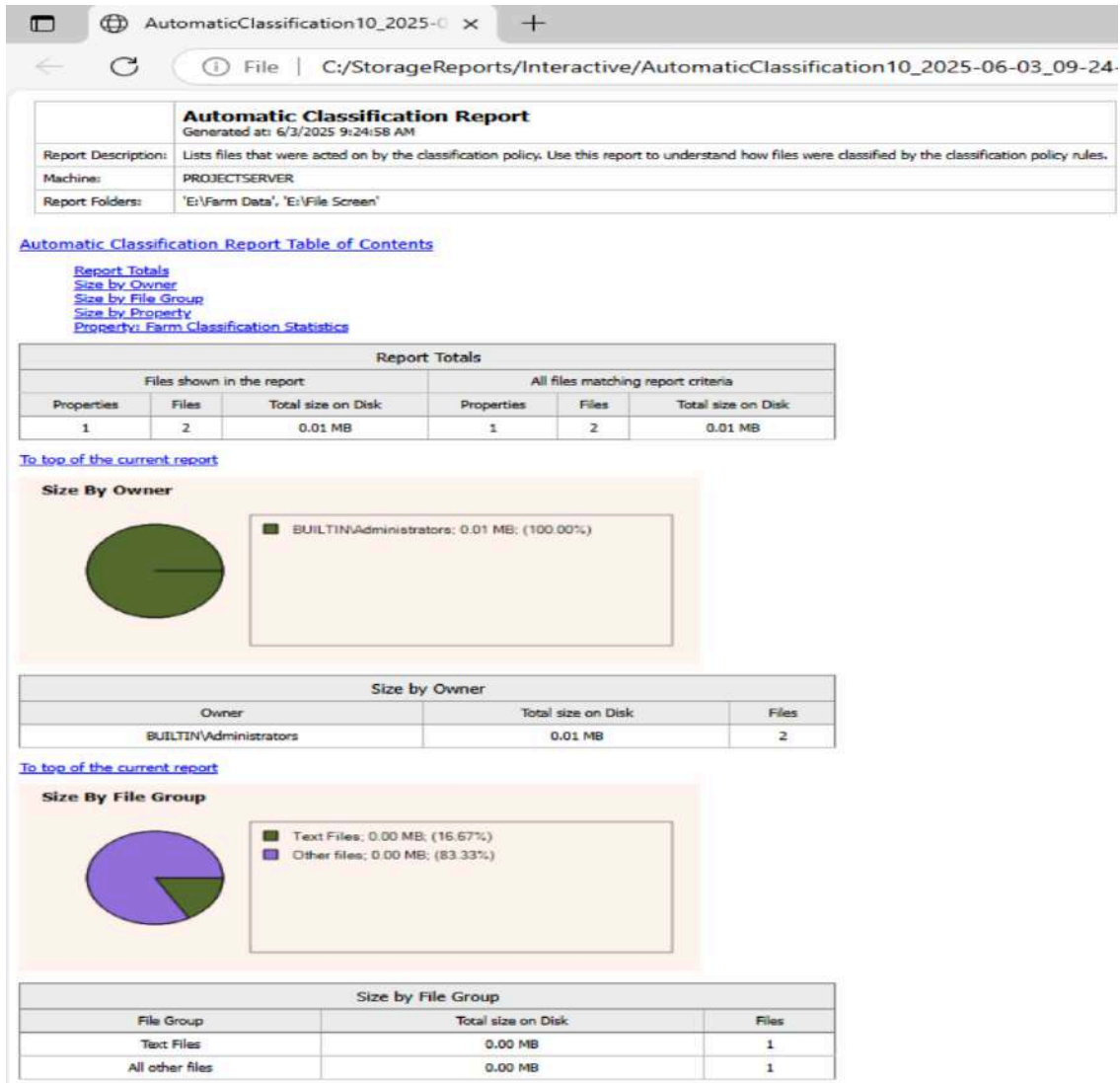
Storage reports configuration for the farm



All the report from the storage management

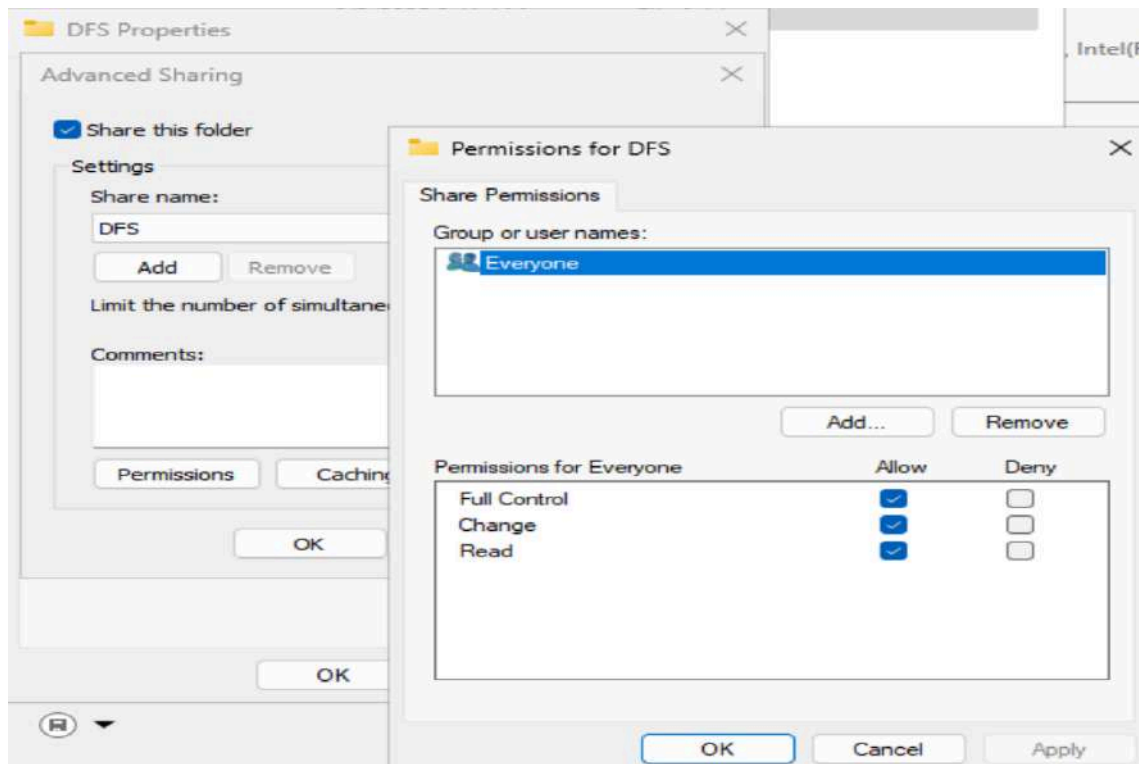
This PC > Windows (C:) > StorageReports > Interactive				
Name	Date modified	Type	Size	
DuplicateFiles2_2025-06-03_09-06-35_files	6/3/2025 9:06 AM	File folder		
FilesbyOwner5_2025-06-03_09-06-35_files	6/3/2025 9:06 AM	File folder		
FilesByType4_2025-06-03_09-06-35_files	6/3/2025 9:06 AM	File folder		
FileScreenAudit3_2025-06-03_09-06-35_files	6/3/2025 9:06 AM	File folder		
LargeFiles6_2025-06-03_09-06-35_files	6/3/2025 9:06 AM	File folder		
LeastRecentlyAccessed7_2025-06-03_09-06-35_files	6/3/2025 9:06 AM	File folder		
MostRecentlyAccessed8_2025-06-03_09-06-35_files	6/3/2025 9:06 AM	File folder		
QuotaUsage9_2025-06-03_09-06-35_files	6/3/2025 9:06 AM	File folder		
DuplicateFiles2_2025-06-03_09-06-35.csv	6/3/2025 9:06 AM	CSV File	1 KB	
DuplicateFiles2_2025-06-03_09-06-35	6/3/2025 9:06 AM	Microsoft Edge H...	6 KB	
DuplicateFiles2_2025-06-03_09-06-35	6/3/2025 9:06 AM	Text Document	5 KB	
DuplicateFiles2_2025-06-03_09-06-35	6/3/2025 9:06 AM	Microsoft Edge H...	3 KB	
FilesbyOwner5_2025-06-03_09-06-35.csv	6/3/2025 9:06 AM	CSV File	2 KB	
FilesbyOwner5_2025-06-03_09-06-35	6/3/2025 9:06 AM	Microsoft Edge H...	15 KB	
FilesbyOwner5_2025-06-03_09-06-35	6/3/2025 9:06 AM	Text Document	12 KB	
FilesbyOwner5_2025-06-03_09-06-35	6/3/2025 9:06 AM	Microsoft Edge H...	16 KB	
FilesByType4_2025-06-03_09-06-35.csv	6/3/2025 9:06 AM	CSV File	2 KB	
FilesByType4_2025-06-03_09-06-35	6/3/2025 9:06 AM	Microsoft Edge H...	10 KB	
FilesByType4_2025-06-03_09-06-35	6/3/2025 9:06 AM	Text Document	7 KB	
FilesByType4_2025-06-03_09-06-35	6/3/2025 9:06 AM	Microsoft Edge H...	6 KB	
FileScreenAudit3_2025-06-03_09-06-35.csv	6/3/2025 9:06 AM	CSV File	1 KB	
FileScreenAudit3_2025-06-03_09-06-35	6/3/2025 9:06 AM	Microsoft Edge H...	3 KB	
FileScreenAudit3_2025-06-03_09-06-35	6/3/2025 9:06 AM	Text Document	3 KB	
FileScreenAudit3_2025-06-03_09-06-35	6/3/2025 9:06 AM	Microsoft Edge H...	2 KB	
LargeFiles6_2025-06-03_09-06-35.csv	6/3/2025 9:06 AM	CSV File	1 KB	
LargeFiles6_2025-06-03_09-06-35	6/3/2025 9:06 AM	Microsoft Edge H...	7 KB	

classification report result:

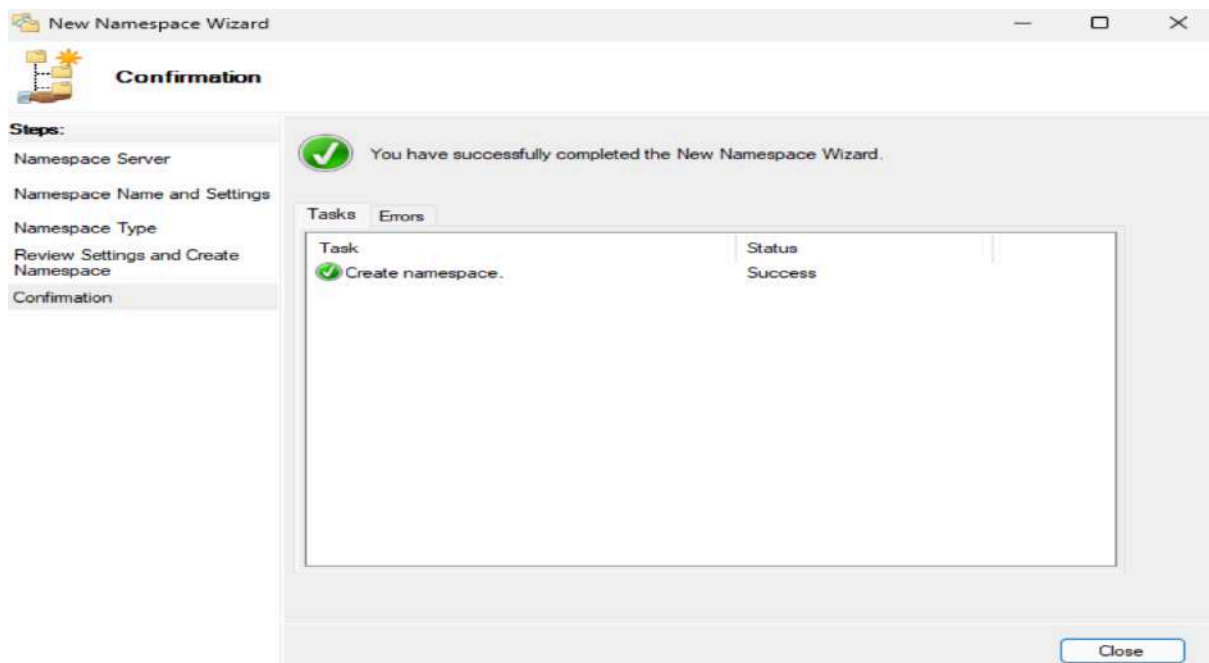


Configuring DFS

Make 1 folder for DFS on each server (main and backup servers)



Create a DFS namespace



there are 2 folder target, one from each server (main and backup)

DFS			
Folder Targets Replication			
2 entries			
Type	Referral Status	Site	Path
	Enabled	Default-First-Site-Name	\\backupproject\DFS
	Enabled	Default-First-Site-Name	\\PROJECTSERVER\DFS\...

Replications for the DFS

Replication

Name

Domain System Volume

farm.local\farmdfs\dfs

testing the DFS from main server to backup server

> Network > BACKUPPROJECT > FarmDFS > DFS >			
Name	Date modified	Type	Size
New folder	6/3/2025 10:19 AM	File folder	
test	6/3/2025 10:21 AM	Text Document	0 KB

testing the DFS from backup server to main server

<< PROJECTSERVER > dfs > Operator >			
Search Operator			
Name	Date modified	Type	Size
New folder	6/3/2025 10:19 AM	File folder	
test	6/3/2025 10:21 AM	Text Document	0 KB

REQUIREMENTS

Hardware : Asus Zenbook Duo

Operating System :

- 1. Windows Server 2025**
- 2. Windows 10 pro**

Software :

- 1. VMWare Workstation Pro**
- 2. Microsoft Word**
- 3. Canva**

Project File Detail		
No.	File Name	Remarks
1.	Implementing Advance Windows Server on E-Agricult	Report
2.	Implementing windows server on E-Agriculture	Presentation
3.		